

Study Guide Introduction to Computational Fluid Dynamics (4RC30) 2025-2026

Lectures

Time: Monday, 8:45-12:30
Thursday, 13:30-17:15
Lecturers: Niels Deen (n.g.deen@tue.nl)
Yali Tang (y.tang2@tue.nl)

Planning

Date	Lecturer	Topic	Study material
April. 20 Mon	Deen	Introduction Conservation equations and diffusion problems	Lecture slides, video recording Chapters 2+4
		Tutorial 1: 1D steady heat transfer problem	Matlab code wc1
April 23 Thu	Tang	Convection-diffusion problems	Lecture slides, videos Chapter 5
		Tutorial 2: 2D steady convection-diffusion problems	Matlab code wc2
April 30 Thu	Deen	Solving PDE's	Video lectures, slides Chapter 6+7
		Tutorial 3: Compressible flow problems	Matlab code wc3
May 7 Thu	Tang	Unsteady problems	Lecture slides, videos Chapter 8
		Tutorial 4: Unsteady flow problems	Matlab code wc4
May 11 Mon	Deen	Turbulence	Video lectures, slides Chapter 3
		Instruction on final assignment: examples, feedback/approval for group flow problems	
May 18 Mon	Deen	Chemical reactions	Lecture slides, videos Chapter 12
		Tutorial 5: Implementation of PDE (combustion)	Matlab code wc5
May 21 Thu	Tang	Boundary conditions	Video lectures, slides Chapter 9
		Tutorial 6: BCs for backward-facing step flow	Matlab code wc6
May 28 Thu	Tang	Outlook on multiphase flow modelling	Lecture slides
		Scientific writing	
Onwards till June 18	Deen & Tang TAs	Group assignment Q&A: Mon (9:00-12:30), Thu (13:30-17:00)	
June 21	Paper deadline		

The book "An Introduction to Computational Fluid Dynamics: The Finite Volume Method" by H.K. Versteeg and W. Malasekera is used as reference material. Lecture slides and recordings are posted on Canvas.

Assessment

The final assignment is to solve a self-defined flow problem in groups of 2 or 3 persons (formed on your own) and write out a paper. The assessment on the group assignment consists of a group grade (80%) and an individual grade (20%). Detailed instruction on assessment will be given during the lectures.